

Enterprise Office Network Using Switch Concepts

Project Title

Design and Implementation of an Enterprise LAN Using Switch Concepts

Project Objective

To understand and implement core Layer 2 switching concepts including:

- Switch operation
- MAC learning and aging
- Frame switching
- Frame flooding
- MAC Address Table management

Using a real-time office network scenario.

Real-Time Scenario

A company called **SNIT Solutions** has opened a new office with:

- HR Department
- Accounts Department
- IT Department
- Training Room

The company wants:

- Fast communication between systems

- Efficient switching
- Minimal network congestion
- Proper learning of device MAC addresses
- Controlled broadcast behavior

You are assigned as a Network Eng to configure and monitor the office switch network.

Because apparently interns are now expected to do production-grade networking while being paid in “experience.”

Network Topology

Devices Used

- 1 Cisco Switch (2960)
 - 8 PCs
 - Ethernet cables
 - Cisco Packet Tracer
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Department Allocation

Department	PC Names	Network
HR	PC1, PC2	192.168.10.0/24
Accounts	PC3, PC4	192.168.10.0/24
IT	PC5, PC6	192.168.10.0/24
Training	PC7, PC8	192.168.10.0/24

IP Addressing

Device	IP Address
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PC1	192.168.10.1
PC2	192.168.10.2
PC3	192.168.10.3
PC4	192.168.10.4
PC5	192.168.10.5
PC6	192.168.10.6
PC7	192.168.10.7
PC8	192.168.10.8

Subnet Mask for all:

255.255.255.0

Concepts Covered in This Project

1. Overview of Switch Concepts

Explanation

A switch connects devices within a LAN and forwards frames using MAC addresses.

Unlike hubs:

- Switches send data only to the destination port
- Reduce collisions
- Improve performance

Real-Time Example

When HR sends payroll data to Accounts:

- Switch checks MAC table

- Sends frame only to Accounts PC
 - Other departments do not receive it
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2. MAC Learning and Aging

Explanation

When a frame enters the switch:

- Switch learns source MAC address
- Stores it in MAC Address Table
- Associates MAC with incoming port

Example:

MAC Address	Port
0001.AAAA.1111	Fa0/1
0001.BBBB.2222	Fa0/4

Aging

If device becomes inactive:

- Switch removes MAC entry after aging timer
- Default aging time: 300 seconds

Real-Time Scenario

Employee laptop disconnects after office hours:

- Switch automatically removes old MAC entry
- Prevents stale entries

Because dead devices hanging around forever is apparently not just a human management problem.

3. Frame Switching

Explanation

When switch receives a frame:

1. Reads destination MAC
2. Checks MAC table
3. Forwards frame to correct port

Example

PC1 sends data to PC4:

- Switch finds PC4 MAC on Fa0/4
- Sends frame only to Fa0/4

This is called **unicast switching**.

4. Frame Flooding

Explanation

If destination MAC is unknown:

- Switch floods frame to all ports except source port

Real-Time Example

New employee connects laptop:

- Switch doesn't know destination MAC yet
- Floods initial frame
- Learns MAC after response

This is called **Unknown Unicast Flooding**.

5. MAC Address Table

Explanation

Switch maintains a table of:

- MAC addresses
- Associated ports
- VLAN information

Command to View Table

show mac address-table

Example Output

Mac Address Table

Vlan	Mac Address	Type	Ports
-----	-----	-----	-----
1	0001.AAAA.1111	DYNAMIC	Fa0/1
1	0001.BBBB.2222	DYNAMIC	Fa0/4

Practical Implementation Steps

Step 1: Create Topology

- Add 1 switch
 - Add 8 PCs
 - Connect using copper straight-through cables
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Step 2: Configure IP Addresses

Assign IP addresses manually to all PCs.

Step 3: Test Connectivity

Use:

ping 192.168.10.4

Observe:

- Initial flooding
- MAC learning
- Faster second ping

Classic networking magic:

First ping fails.

Second ping works.

Everyone pretends this is emotionally acceptable behavior.

Step 4: Verify MAC Learning

On switch CLI:

enable

show mac address-table

Observe learned MAC entries.

Step 5: Test MAC Aging

Wait for aging timer or clear manually:

clear mac address-table dynamic

Send traffic again and observe relearning.

Expected Outcomes

After completing project, intern will understand:

- How switches forward frames
 - MAC learning process
 - Unknown unicast flooding
 - MAC address table operation
 - Aging mechanism
 - Enterprise LAN communication
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Tools Required

Tool	Purpose
Cisco Packet Tracer	Network simulation
Cisco 2960 Switch	Switching
PCs	End devices

Advanced Add-On (Optional)

You can extend project by adding:

- VLANs
- Port Security
- STP
- Inter-VLAN Routing

This makes the project look far more “industry-level” on internship reports. Recruiters enjoy buzzwords almost as much as switches enjoy blinking lights.

Sample Viva Questions

1. What is MAC learning?
 2. What is frame flooding?
 3. Difference between switch and hub?
 4. What is MAC aging?
 5. Why does a switch maintain MAC table?
 6. What happens when destination MAC is unknown?
 7. What command displays MAC table?
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Cisco Exam Question

Which function forwards frames to ports that have a matching destination MAC address?

- A. frame flooding
- B. frame filtering
- C. frame pushing
- D. frame switching

What is the default behavior of a Layer 2 switch when a frame with an unknown destination MAC address is received?

- A. The Layer 2 switch forwards the packet and adds the destination MAC address to its MAC address table.
- B. The Layer 2 switch sends a copy of a packet to CPU for destination MAC address learning.
- C. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN.
- D. The Layer 2 switch drops the received frame.

Correct Answer: **C. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN.**

Explanation:

When a switch receives a frame and does not know the destination MAC address:

1. It checks the MAC address table.
2. Destination MAC is not found.
3. Switch performs **unknown unicast flooding**.
4. The frame is sent out all ports in the same VLAN except the port where it was received.

Because apparently networking equipment handles uncertainty by yelling the message at everyone in the room and hoping the right device answers. Ancient village communication strategy, now in silicon form.

Real-Time Example:

- PC1 sends data to PC2.
- Switch already knows PC1 MAC address but not PC2 MAC.
- Switch floods the frame to all ports in VLAN 10 except PC1's port.
- PC2 receives it and replies.
- Switch learns PC2 MAC address and stores it in MAC table.
- Future traffic goes directly to PC2 port only.

Important Concept:

- **Known MAC Address** → Frame Switching (direct forwarding)
- **Unknown MAC Address** → Flooding
- **Broadcast MAC (FF:FF:FF:FF:FF:FF)** → Broadcast Flooding
- **Wrong VLAN** → Frame dropped

What is a characteristic of frame switching?

- A. rewrites the source and destination MAC address
- B. performs a lookup to learn the destination interface
- C. sends a retransmission request when a new frame is received
- D. inspects and drops frames from unknown destinations

Correct Answer: **B. performs a lookup to learn the destination interface**

Explanation:

In frame switching, a switch:

1. Receives the Ethernet frame.
2. Reads the destination MAC address.
3. Checks the MAC address table (CAM table).
4. Finds the correct outgoing interface/port.
5. Forwards the frame only to that port.

Because apparently every switch spends its life doing instant address-book lookups faster than humans can find their own contacts.

Why other options are wrong:

- **A. rewrites the source and destination MAC address**
Routers rewrite MAC addresses when forwarding between networks, not normal Layer 2 switching.
- **C. sends a retransmission request when a new frame is received**
Ethernet switches do not request retransmissions like this.
- **D. inspects and drops frames from unknown destinations**
Unknown destination frames are usually flooded, not dropped.

Real-Time Example:

- PC-A sends data to Server-A.
- Switch receives frame.
- Destination MAC = 00:1A:2B:3C:4D:5E
- Switch checks MAC table.
- Finds MAC connected to Port 10.
- Frame forwarded only to Port 10.

How does MAC learning function?

- A. restricts ports to a maximum of 10 dynamically-learned addresses
- B. increases security on the management VLAN
- C. drops received MAC addresses not listed in the address table
- D. associates the MAC address with the port on which it is received

Correct Answer: **D. associates the MAC address with the port on which it is received**

Explanation:

MAC learning is the process where a switch learns which devices are connected to which ports.

When a frame arrives:

1. The switch reads the **source MAC address**.
2. It records that MAC address in the MAC/CAM table.
3. It associates the MAC address with the incoming port.

So later, when another device sends traffic to that MAC address, the switch knows exactly where to forward the frame. Tiny digital receptionist maintaining a seating chart nobody asked for but everyone depends on.

Real-Time Example:

- PC1 connected to Port Fa0/1
MAC = AA:AA:AA:AA:AA:AA
- PC1 sends data.
- Switch learns:

AA:AA:AA:AA:AA:AA → Fa0/1

Later:

- PC2 wants to send data to PC1.
- Switch checks MAC table.
- Frame forwarded directly to Fa0/1.

Why other options are wrong:

- **A** → This refers to port security limits, not MAC learning.
- **B** → MAC learning itself does not increase VLAN security.
- **C** → Unknown MAC addresses are usually flooded, not dropped.

Key Concept:

- **Source MAC address** → Learned
- **Destination MAC address** → Looked up for forwarding

Switch behavior in one sentence:

“I saw this MAC on this port before, so that’s where I’ll send traffic.”

Conclusion

This project demonstrates how Layer 2 switches intelligently forward frames in an enterprise LAN using MAC-based forwarding techniques. The implementation improves communication efficiency, reduces unnecessary traffic, and provides practical understanding of switching operations used in real corporate environments.